

Sathishkumar Thirumalai

[LinkedIn](#) | [GitHub](#) | [Kaggle](#)

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AWS CERTIFIED — SENIOR LEAD DATA SCIENTIST — AI SOLUTIONS ARCHITECT

14+ years of systems engineering experience, including 10+ years delivering enterprise Machine Learning, Generative AI, and Agentic AI solutions across AWS, GCP, and on-prem environments. Led teams that delivered production-grade AI systems to automate complex workflows, reduce operational costs, and scale business operations. Implemented organisation-wide AI frameworks and strategies for improving efficiency of operations. Proven Impact delivered using Agentic AI, Multi-Agent Systems, LLM Applications, Computer Vision, Forecasting, Anomaly Detection, and Classical Machine Learning.

SKILLS

Languages	: Python, C#, C++, C, PyQt, Rest APIs, Flask, FastAPI
ML/DL/AI	: Machine Learning, Generative AI, Agentic AI, RAG, Computer Vision, Forecasting, Anomaly , Detection, MLOps
Data Tools	: Databricks, Apache Airflow, Apache Kafka, AWS Glue, AWS Sagemaker DataWrangler, Google PubSub, PySpark, numpy, pandas, MSSQL, SQLite, FAISS, ChromaDB, matplotlib, seaborn
ML/DL Tools	: scikit-learn, PyTorch, Tensorflow Keras, OpenCV, Hugging Face, MLflow
AI/LLM Tools	: LangChain, LangGraph, Google ADK, agno-ai, Model Context Protocol (MCP), Ollama, Lang-fuse
Cloud Tools	: AWS (Bedrock, SageMaker, EKS), GCP (Vertex AI, Agent Engine)

SELECTED ENTERPRISE AI SYSTEMS

Census Processing Agent (AWS), Forecasting Platform(AWS), Enterprise RAG Platform(GCP), ANPR Platform(On-prem)

EXPERIENCE

Senior Lead Data Scientist

Guardian Life India

Feb 2026 – present

Chennai, India

- **Census Processing Agent:** Led cross-functional delivery of Guardian's first Agentic AI solution for census processing, improving accuracy from 40%→93%, reducing Average Handling Time from 45 minutes to 2 minutes, increasing operational throughput by approximately 10× and reducing costs by upto 20×. Delivered the solution two quarters ahead of schedule. **Stack:** [Python, Claude Sonnet, AWS Bedrock, AWS EKS]
- **Group Behaviour Modelling:** Engineered a predictive model to predict the groups that would go into Due & Unpaid list. Performed data extraction and transformation, feature engineering and model training. Performed SHAPley analysis to determine the root cause and provide actionable insights for the business. Achieved reduction in volume of unpaid bills. **Stack:** [Python, XGBoost, SHAP, FastAPI]

Senior Data Scientist

Indira Gandhi Centre for Atomic Research (IGCAR) — Technical Officer

Nov 2012– Feb 2026

Chennai, India

- **On-prem deployment: System Failure Forecasting :** Engineered real-time forecasting and anomaly detection system that pre-empted 92 % of system failures annually **Stack:** [MS-SQL, Apache Kafka, Apache Airflow, Python, pandas, Prophet, FastAPI]
- **On-prem deployment: Automatic Number Plate Recognition :** Architected and developed a production-grade real-time ANPR solution, achieving over 95% vehicle recognition accuracy through computer vision. **Stack:**[OpenCV, Object Detection, TrOCR, Python, MS-SQL, Gradio]
- **On-prem deployment: Camouflaged Object Detection :** Architected and trained a computer vision CNN model on COD-10K dataset to segment camouflaged objects. **Stack:**[OpenCV, PyTorch, Weights and Biases, MLFlow]

- **GCP Deployment: (Generative AI) RAG-QA Agent** : Architected, developed and deployed a RAG-QA agent powered by Gemini 2.5-pro and GCP Vertex AI Search to answer questions about biometric access control systems. Built automated pipelines to continuously update the knowledge base, deployed the agent onto GCP. **Stack**: [Scrapy, ChromaDB, FAISS, Python, Google-GenAI, Google ADK, Vertex AI Search, Google Agent-Engine, Google Cloud Monitoring]
- **On-prem deployment: (Generative AI) Analytics Agent** : Built a production analytics agent using Llama 3.1 8B and function calling to query and analyze MS SQL transaction data. **Stack**: [MS-SQL, Ollama, LangChain, LangGraph, Python, MCP Server, Gradio]
- **Multi-source ETL Pipelines** : Created automatic ETL pipelines for user enrollment into biometric systems. **Stack**: [Apache Kafka, Apache Airflow, Python, MS-SQL, RESTAPI]
- **C, C# Application Development** : Designed and Developed an enterprise level physical access control system. **Stack** : [Embedded C, C#, MS-SQL]
- **Python Application Development** : Designed and Developed a Photo Display System to read RFID cards and display photos. **Stack** : [Python, PyQt, SQLite, Apache Kafka]
- **Algorithm Design** : Researched, implemented, and optimized biometric recognition algorithms — including iris template matching and cancellable fingerprint templates **Stack** : [OpenCV, Python, SQLite, C#]

DotNet Developer

Infosys — Systems Engineer

Aug 2011 – Nov 2012

Chennai, India

- **DotNet Developer** : Underwent intense on-prem training on DotNet technologies.

EDUCATION

Sri Siva Subramaniya Nadar College of Engineering

Bachelor of Engineering in Electronics and Communication

2007 – 2011

Chennai, Tamil Nadu, India

Homi Bhabha National Institute (HBNI)

Master of Science (Engg) in Computer Science (Algorithm Design)

2015 – 2018

Chennai, Tamil Nadu, India

OPEN-SOURCE CONTRIBUTIONS

Telegmail : A Python app for real-time forwarding of messages from telegram channel to gmail [Source Code](#)

Nutrition info : Powered by gemini 1.5-flash for giving nutritional information on Indian foods [Source Code](#)

Genai-calendar-assistant : Agent powered by gemini 2.0-flash for assisting in google calendar tasks [Source Code](#)

COMPETITIONS

- 2025: Top **15 out of 9000** finish @ Agentic AI day Hackathon organized by Google Cloud. Built an application for managing city data overload. **Stack** : [Google CloudShell, Google PubSub, GoogleMapsAPI, Google ADK, Cloud Functions, GCP Agent Engine] [Source Code](#)
- 2025: **SwagWinner** @ Agents-MCP-Hackathon organized by HuggingFace. Built a MCP server that contains tools for evaluating RAG systems and deployed on HuggingFace spaces **Stack** : [Python, BM25, HuggingFace, Gradio, MCP Server] [Source Code](#)
- 2025: **Second Place** @ DeHaDo-AI competition organized by IIT Jammu. Architected and trained a handwritten document recognition pipeline with Object Detection and Transformer models. **Stack** : [Python, OpenCV, YOLO, TrOCR, PyTorch, MLFlow] [Source Code](#)
- 2025: **13th place prize money winner** @ Innovate with Government of India Data Visualization Hackathon organized by Ministry of Statistics and Programme Implementation (MoSPI). Analyzed publicly available government data, make visualizations and offered insights. **Stack** : [Python, pandas, matplotlib, seaborn] [Source Code](#)

CERTIFICATIONS

AWS Machine Learning Engineer- Associate (valid until 27-09-2028)